

Usare Windows Powershell

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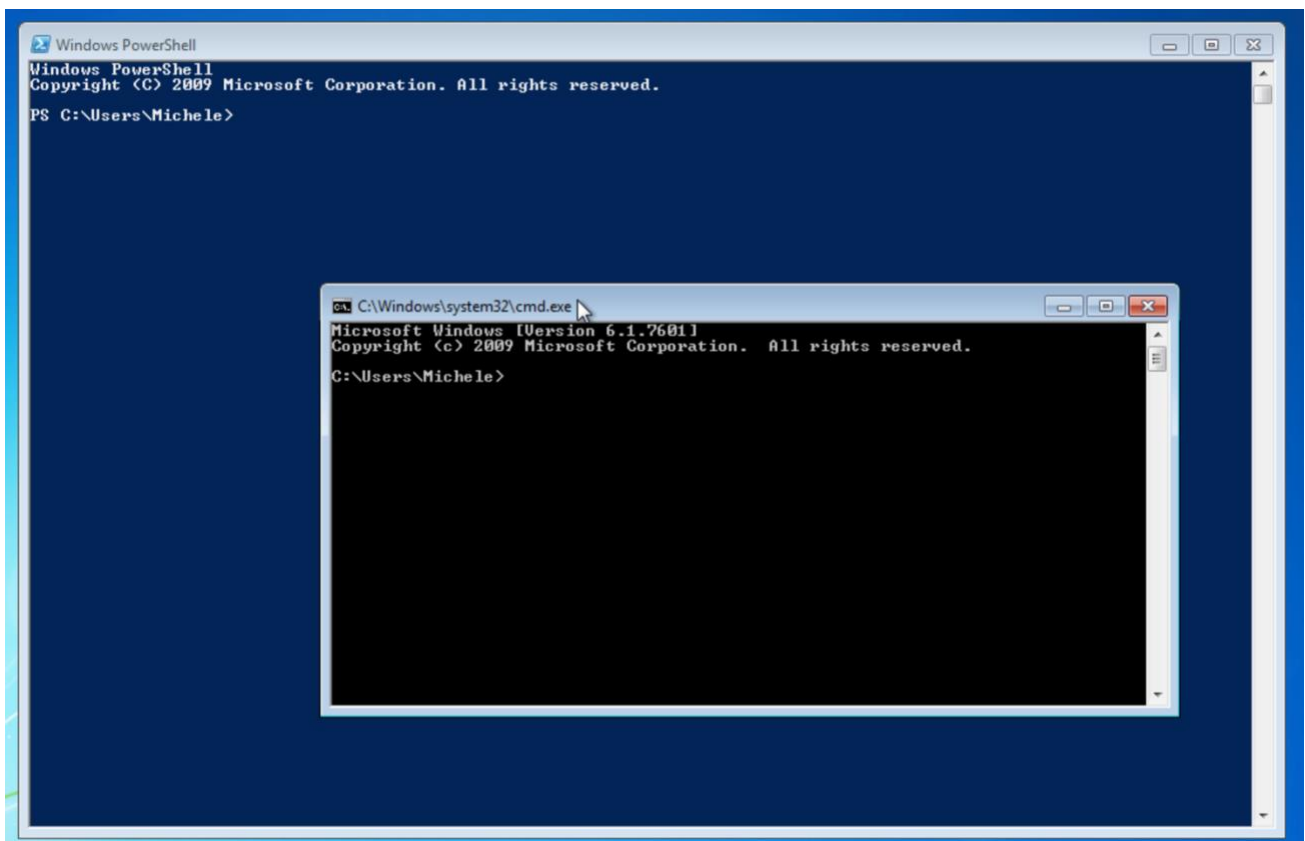
1 Scopo del progetto

L'obiettivo del laboratorio è esplorare alcune funzioni di PowerShell. Durante l'esercizio vengono analizzate l'apertura della console, l'uso dei comandi del Prompt dei comandi e di PowerShell, l'esplorazione dei cmdlet, l'utilizzo di netstat e lo svuotamento del cestino.

2 Svolgimento

2.1 Accesso alla PowerShell e Prompt

Per aprire PowerShell, bisogna fare clic su Start, cercare PowerShell e selezionarlo. Clicchiamo anche su il Prompt dei comandi cercandolo sempre dal menu Start.

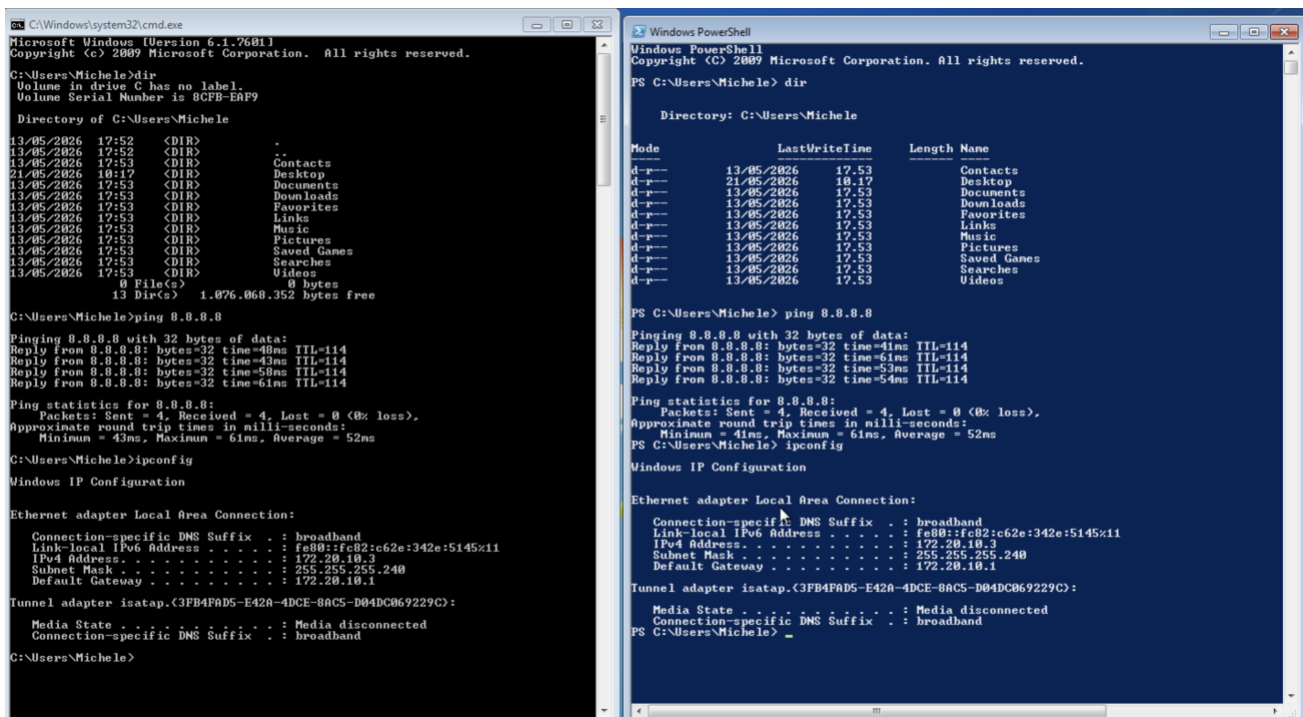


2.2 Inserimento comandi

Inserendo dir nel Prompt dei comandi, il sistema mostra l'elenco dei file e delle cartelle presenti nella directory corrente, insieme a informazioni come data, ora e dimensione. In PowerShell, lo stesso comando produce un risultato identico.

Usando altri comandi nel Prompt dei comandi o in PowerShell si ottengono risultati uguali a seconda del comando.

- **ping** verifica la raggiungibilità di un host sulla rete e mostra il tempo di risposta.
- **cd** serve per cambiare directory.
- **ipconfig** mostra le informazioni di configurazione della rete, come indirizzo IP e gateway.



```
Microsoft Windows [Version 6.1.7601]
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C:\Users\Michele>dir
Volume in drive C has no label.
Volume Serial Number is 8CFB-EAF9

Directory of C:\Users\Michele

13/05/2026 17:52 <DIR> .
13/05/2026 17:52 <DIR> ..
13/05/2026 17:53 <DIR> Contacts
21/05/2026 10:17 <DIR> Desktop
13/05/2026 17:53 <DIR> Documents
13/05/2026 17:53 <DIR> Downloads
13/05/2026 17:53 <DIR> Favorites
13/05/2026 17:53 <DIR> Links
13/05/2026 17:53 <DIR> Music
13/05/2026 17:53 <DIR> Pictures
13/05/2026 17:53 <DIR> Saved Games
13/05/2026 17:53 <DIR> Searches
13/05/2026 17:53 <DIR> Videos
               0 File(s)      0 bytes
               13 Dir(s)      1.076.068.352 bytes free

C:\Users\Michele>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=43ms TTL=114
Reply from 8.8.8.8: bytes=32 time=43ms TTL=114
Reply from 8.8.8.8: bytes=32 time=58ms TTL=114
Reply from 8.8.8.8: bytes=32 time=61ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 43ms, Maximum = 61ms, Average = 52ms

C:\Users\Michele>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : broadband
    Link-local IPv6 Address . . . . . : fe80::fc82:c62e:342e:5145x11
    IPv4 Address. . . . . : 172.20.10.3
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : 172.20.10.1

Tunnel adapter isatap.{3FB4FAD5-E42A-4DCE-8AC5-D04DC869229C}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : broadband

C:\Users\Michele>

Windows PowerShell
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PS C:\Users\Michele> dir

Directory: C:\Users\Michele

Mode                LastWriteTime         Length Name
----                -
d-r--             13/05/2026         17.53      Contacts
d-r--             21/05/2026         10.17      Desktop
d-r--             13/05/2026         17.53      Documents
d-r--             13/05/2026         17.53      Downloads
d-r--             13/05/2026         17.53      Favorites
d-r--             13/05/2026         17.53      Links
d-r--             13/05/2026         17.53      Music
d-r--             13/05/2026         17.53      Pictures
d-r--             13/05/2026         17.53      Saved Games
d-r--             13/05/2026         17.53      Searches
d-r--             13/05/2026         17.53      Videos

PS C:\Users\Michele> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=41ms TTL=114
Reply from 8.8.8.8: bytes=32 time=61ms TTL=114
Reply from 8.8.8.8: bytes=32 time=53ms TTL=114
Reply from 8.8.8.8: bytes=32 time=54ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 41ms, Maximum = 61ms, Average = 52ms

PS C:\Users\Michele> ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

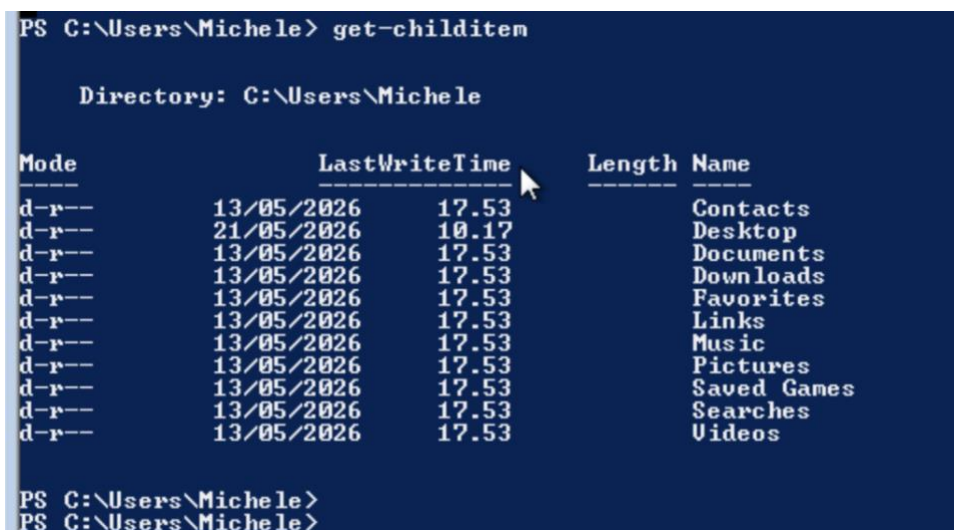
    Connection-specific DNS Suffix  . : broadband
    Link-local IPv6 Address . . . . . : fe80::fc82:c62e:342e:5145x11
    IPv4 Address. . . . . : 172.20.10.3
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : 172.20.10.1

Tunnel adapter isatap.{3FB4FAD5-E42A-4DCE-8AC5-D04DC869229C}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : broadband

PS C:\Users\Michele>
```

Un comando equivalente a dir in PowerShell è Get-Childitem



```
PS C:\Users\Michele> get-childitem

Directory: C:\Users\Michele

Mode                LastWriteTime         Length Name
----                -
d-r--             13/05/2026         17.53      Contacts
d-r--             21/05/2026         10.17      Desktop
d-r--             13/05/2026         17.53      Documents
d-r--             13/05/2026         17.53      Downloads
d-r--             13/05/2026         17.53      Favorites
d-r--             13/05/2026         17.53      Links
d-r--             13/05/2026         17.53      Music
d-r--             13/05/2026         17.53      Pictures
d-r--             13/05/2026         17.53      Saved Games
d-r--             13/05/2026         17.53      Searches
d-r--             13/05/2026         17.53      Videos

PS C:\Users\Michele>
PS C:\Users\Michele>
```

2.3 Esplorare il comando Netstat

Con il comando netstat -h ho visualizzato le opzioni disponibili, mentre con netstat -r ho ottenuto la tabella di routing. Dal risultato è emerso che il gateway IPv4 è 172.20.10.1

```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32> netstat -h

Displays protocol statistics and current TCP/IP network connections.

NETSTAT [-a] [-b] [-e] [-f] [-n] [-o] [-p proto] [-r] [-s] [-t] [interval]

-a          Displays all connections and listening ports.
-b          Displays the executable involved in creating each connection or
           listening port. In some cases well-known executables host
           multiple independent components, and in these cases the
           sequence of components involved in creating the connection
           or listening port is displayed. In this case the executable
           name is in [i] at the bottom, on top is the component it called,
           and so forth until TCP/IP was reached. Note that this option
           can be time-consuming and will fail unless you have sufficient
           permissions.
-e          Displays Ethernet statistics. This may be combined with the -s
           option.
-f          Displays Fully Qualified Domain Names (FQDN) for foreign
           addresses.
-n          Displays addresses and port numbers in numerical form.
-o          Displays the owning process ID associated with each connection.
-p proto    Shows connections for the protocol specified by proto; proto
           may be any of: TCP, UDP, ICMPv6, or UDPv6. If used with the -s
           option to display per-protocol statistics, proto may be any of:
           IP, IPv6, ICMP, ICMPv6, TCP, TCPv6, UDP, or UDPv6.
-r          Displays the routing table.
-s          Displays per-protocol statistics. By default, statistics are
           shown for IP, IPv6, ICMP, ICMPv6, TCP, TCPv6, UDP, and UDPv6;
           the -p option may be used to specify a subset of the default.
-t          Displays the current connection offload state.
interval    Redisplay selected statistics, pausing interval seconds
           between each display. Press CTRL+C to stop redisplaying
           statistics. If omitted, netstat will print the current
           configuration information once.

PS C:\Windows\system32> netstat -r

=====
Interface List
11...a2 f4 fe a4 e8 55 .....Realtek RTL8139C+ Fast Ethernet NIC
1.....Software Loopback Interface 1
12...00 00 00 00 00 00 e0 Microsoft ISATAP Adapter
=====
```

```
PS C:\Windows\system32> netstat -r

=====
Interface List
11...a2 f4 fe a4 e8 55 .....Realtek RTL8139C+ Fast Ethernet NIC
1.....Software Loopback Interface 1
12...00 00 00 00 00 00 e0 Microsoft ISATAP Adapter
=====

IPv4 Route Table
=====
Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
0.0.0.0                0.0.0.0          172.20.10.1      172.20.10.3      20
127.0.0.0              255.0.0.0        On-link          127.0.0.1        306
127.0.0.1              255.255.255.255  On-link          127.0.0.1        306
127.255.255.255        255.255.255.255  On-link          127.0.0.1        306
172.20.10.0            255.255.255.240  On-link          172.20.10.3      276
172.20.10.3            255.255.255.255  On-link          172.20.10.3      276
172.20.10.15           255.255.255.255  On-link          172.20.10.3      276
224.0.0.0              240.0.0.0        On-link          127.0.0.1        306
224.0.0.0              240.0.0.0        On-link          172.20.10.3      276
255.255.255.255        255.255.255.255  On-link          127.0.0.1        306
255.255.255.255        255.255.255.255  On-link          172.20.10.3      276
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination Gateway
1 306 ::1/128 On-link
11 276 fe80::/64 On-link
11 276 fe80::fc82:c62e:342e:5145/128 On-link
1 306 ff00::/8 On-link
11 276 ff00::/8 On-link
=====
Persistent Routes:
None
PS C:\Windows\system32>
```

Dalla scheda Dettagli di Gestione Attività si possono ottenere il nome del processo, il PID, lo stato, l'utente, l'uso di CPU e memoria e la descrizione del processo. Dalla finestra Proprietà si possono visualizzare ulteriori informazioni utili per identificare il processo selezionato.

Administrator: Windows PowerShell

```

[svchost.exe]
UDP 0.0.0.0:5004 ***
[wmipnetk.exe]
UDP 0.0.0.0:5005 ***
[wmipnetk.exe]
UDP 0.0.0.0:5355 ***
Dnscache
[svchost.exe]
UDP 0.0.0.0:49320 ***
FDResPub
[svchost.exe]
UDP 127.0.0.1:1900 ***
SSDPSRV
[svchost.exe]
UDP 127.0.0.1:50018 ***
SSDPSRV
[svchost.exe]
UDP 172.20.10.3:137 ***
Can not obtain ownership information
UDP 172.20.10.3:138 ***
Can not obtain ownership information
UDP 172.20.10.3:1900 ***
SSDPSRV
[svchost.exe]
UDP [::]:13702 ***
FDResPub
[svchost.exe]
UDP [::]:13702 ***
FDResPub
[svchost.exe]
UDP [::]:5004 ***
[wmipnetk.exe]
UDP [::]:5005 ***
[wmipnetk.exe]
UDP [::]:5355 ***
Dnscache
[svchost.exe]
UDP [::]:49321 ***
FDResPub
[svchost.exe]
UDP [::]:1900 ***
SSDPSRV
[svchost.exe]
UDP [::]:50017 ***
SSDPSRV
[svchost.exe]
UDP [fe80::fc82:c62e:342e:5145%11]:1900 ***
SSDPSRV
[svchost.exe]
PS C:\Windows\system32>

```

Windows Task Manager

File Options View Help

Applications Processes Services Performance Networking Users

Name	PID	Description	Status	Group
SENS	784	System Ev...	Runn...	netsvcs
Schedule	784	Task Sched...	Runn...	netsvcs
ProfSvc	784	User Profil...	Runn...	netsvcs
LanmanServer	784	Server	Runn...	netsvcs
iphlpvc	784	IP Helper	Runn...	NetSvcs
gpsvc	784	Group Polic...	Runn...	netsvcs
BITS	784	Backgroun...	Runn...	netsvcs
Appinfo	784	Application...	Runn...	netsvcs
AeLookupSvc	784	Application...	Runn...	netsvcs
WdiSystem...	756	Diagnostic ...	Runn...	LocalSyste...
UxSms	756	Desktop Wi...	Runn...	LocalSyste...
TrkWks	756	Distributed...	Runn...	LocalSyste...
SysMain	756	Superfetch	Runn...	LocalSyste...
PcaSvc	756	Program C...	Runn...	LocalSyste...
Netman	756	Network C...	Runn...	LocalSyste...
CscService	756	Offline Files	Runn...	LocalSyste...
AudioEnda...	756	Windows A...	Runn...	LocalSyste...
wscsv	720	Security C...	Runn...	LocalServic...
lmhosts	720	TCP/IP Net...	Runn...	LocalServic...
eventlog	720	Windows E...	Runn...	LocalServic...
Dhcp	720	DHCP Client	Runn...	LocalServic...
AudioSrv	720	Windows A...	Runn...	LocalServic...
RpcSs	656	Remote Pr...	Runn...	rpcss
RpcEptMap...	656	RPC Endpo...	Runn...	RPCSS
Spooler	644	Print Spooler	Runn...	N/A
Power	584	Power	Runn...	DcomLaunch
PlugPlay	584	Plug and Play	Runn...	DcomLaunch

Processes: 36 CPU Usage: 0% Physical Memory: 11%

2.4 Svuotare il cestino con PowerShell

Il comando `clear-recyclebin` ha chiesto conferma e, dopo l'accettazione, ha svuotato il Cestino eliminando tutti i file presenti.

```
Administrator: Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator>
PS C:\Users\Administrator>
PS C:\Users\Administrator> clear-recyclebin

Confirm
Are you sure you want to perform this action?
Performing the operation "Clear-RecycleBin" on target "All of the contents of the Recycle Bin".
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"): y
PS C:\Users\Administrator> |
```

2.5 Analisi comandi aggiuntivi

Per un analista di sicurezza, PowerShell può essere utile con comandi come `Get-Process`, `Get-Service`, `Get-NetTCPConnection` e `Get-WinEvent`.

```
ProviderName: Microsoft-Windows-CertificationAuthority

TimeCreated          Id LevelDisplayName Message
-----
19/06/2026 11:09:56   26 Information Active Directory Certificate Services for Mordor-MORDORSERVER-C...

ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated          Id LevelDisplayName Message
-----
19/06/2026 11:09:56   4634 Information An account was logged off...
19/06/2026 11:09:56   4624 Information An account was successfully logged on...
19/06/2026 11:09:56   4672 Information Special privileges assigned to new logon...
19/06/2026 11:09:56   4648 Information A logon was attempted using explicit credentials....
PS C:\Users\Administrator> get-nettcpconnection

LocalAddress          LocalPort RemoteAddress          RemotePort State      AppliedSetting
-----
::                   49882    ::                      0          Bound
::                   49881    ::                      0          Bound
::                   49880    ::                      0          Bound
::                   49879    ::                      0          Bound
::                   49760    ::                      0          Bound
::                   49748    ::                      0          Bound
::                   49746    ::                      0          Bound
::                   49721    ::                      0          Bound
::                   49719    ::                      0          Bound
::                   49710    ::                      0          Bound
```

Per implementare filtri nell'analisi dei log in PowerShell posso usare `Select-String` per i file di testo e `Get-WinEvent -FilterHashtable` per i log di Windows. In questo modo posso filtrare per livello, ID evento, sorgente, messaggio o intervallo di tempo, rendendo l'analisi più veloce e precisa.

3 Conclusioni

L'esercizio ha permesso di familiarizzare con l'ambiente Windows PowerShell e con i principali comandi utili per l'amministrazione del sistema.

Un altro aspetto importante è stato l'uso di PowerShell in ambito sicurezza informatica. Grazie ai comandi di filtraggio e di analisi dei log, è possibile raccogliere informazioni utili per individuare attività sospette, controllare processi, servizi e connessioni di rete, e rendere più veloce l'analisi di un sistema.